Eric Fontes

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Education

University of Florida | Gainesville, FL *Aug 2022 — May 2026*

- B.S. Computer Science, B.S. Mathematics (GPA: 3.81/4.0)
- *Relevant Coursework*: Python, Statistics, Fundamentals of Machine Learning, Linear Algebra, Fourier Analysis, Differential Equations, Sets and Logic, Data Structures and Algorithms
- *Certifications*: AI Fundamentals and Applications, NVIDIA Fundamentals of Deep Learning

Experience

Altru, *Software Engineering Intern* | Remote & Miami, FL *July 2026 — Current*

- Led a database security audit that **surfaced 18 vulnerabilities and remediated 11**, proving **4** Postgres RLS gaps live-exploitable
- Cut **upload payloads by 27%** via image encoding and video compression, and improved app responsiveness by **removing 25%** of redundant screen refreshes

Metis, *Student Project Backend Engineer* | Gainesville, FL *January 2026 — May 2026*

- Evaluated retrieval quality across dense semantic search and BM25 keyword search configurations, selecting a hybrid approach that achieved **34% higher Recall@5** over single-method baselines
- Built a FastAPI backend as a provider-agnostic LLM gateway across **Anthropic, OpenAI, and OpenRouter**, enabling model selection without code changes
- Assessed tool-use outcomes for a streaming ReAct-based agent integrating Tavily-powered web search within an agentic AI workflow

Capital One, *Tech Mini Mester Participant* | Remote *July 2024 — August 2024*

- **1 out of 70** exclusively selected to participate in a 5-day immersive experience to enhance technical and soft skills
- Built *devfetch*, a CLI-based project, and **presented to Capital One leadership**, earning a **top 3 project recognition**

The Mixtape Engine, *Student Project Lead* | Gainesville, FL *March 2024 — May 2024*

- Processed **114,000+ rows** of song records to build a recommendation pipeline, custom playlist generation in **under 150ms per query**
- Improved algorithm runtime by **3.2x** and reduced memory usage by **44%** in merge sort implementation
- Integrated with Spotify's REST API to fetch album art and display song metadata to frontend
- Led a 4-person development team, coordinating weekly standups and managed Git workflows

Skills

- **Languages**: Python, C++, C, Java, Rust, MATLAB, Typescript, HTML/CSS, Bash
- **Frameworks & Libraries**: React, NextJS, Node.js, Svelte, Tauri, FastAPI, PyTorch, pandas, NumPy
- **Developer Tools**: Git, Github, Docker, Linux, UNIX

Projects

Metis, *AI-powered research paper assistant* *January 2026 — May 2026*

- Architected a hybrid retrieval pipeline combining dense semantic search and BM25 keyword search within a full-stack desktop application (Tauri + React) for ingesting and querying academic PDFs
- Abstracted LLM calls behind a FastAPI-served, provider-agnostic interface spanning **Anthropic, OpenAI, and OpenRouter**
- Extended document ingestion with multimodal OCR (pix2text) to support scanned and image-based PDFs
- Implemented a high-performance PDF rendering engine in Rust with **under 150ms page render latency**